



IIT Madras
B.S. Degree

PRESENTATION SKILLS



*Strategies for
Professional Growth*

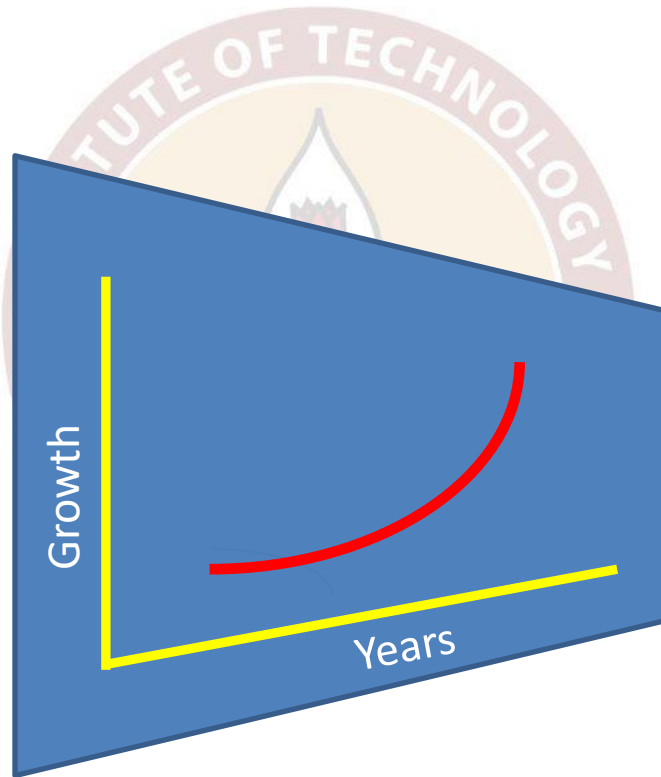
WEEK

3

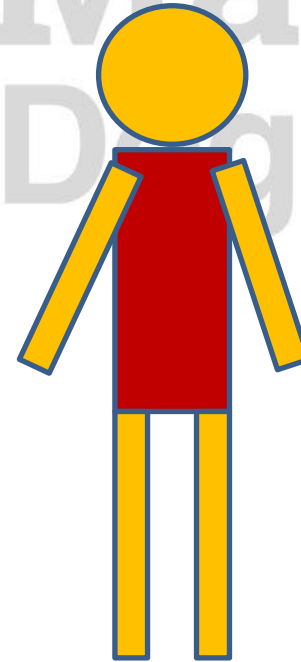


PRATHAP HARIDOSS
PROFESSOR, IIT MADRAS

How to make a Presentation



IIT Madras
B.S. Degree



Outline:

- Professional presentation - setting
- Constraints ←
- Connecting with your audience ←
- Illustrations ←
- Summary

What to do and what not to do!



Outline:

- **Professional presentation - setting**
- Constraints
- Connecting with your audience
- Illustrations
- Summary

Professional Presentation - Setting:

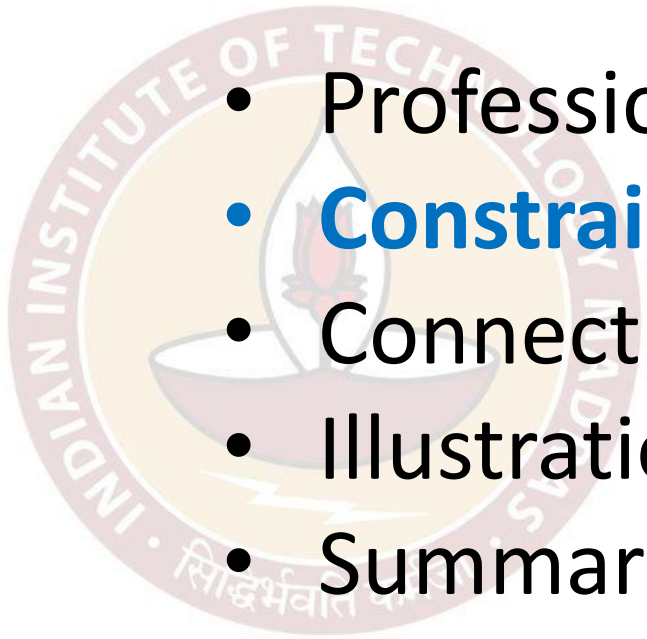
- Primarily a “serious” activity
- Likely you are being evaluated ←
- Reporting a status ←
- Conveying an idea ←

Professional Presentation - Setting:

- Is an event that can be repeated ←
- Structure based on organization or event ←
- Interactive
- Specific audience each occasion

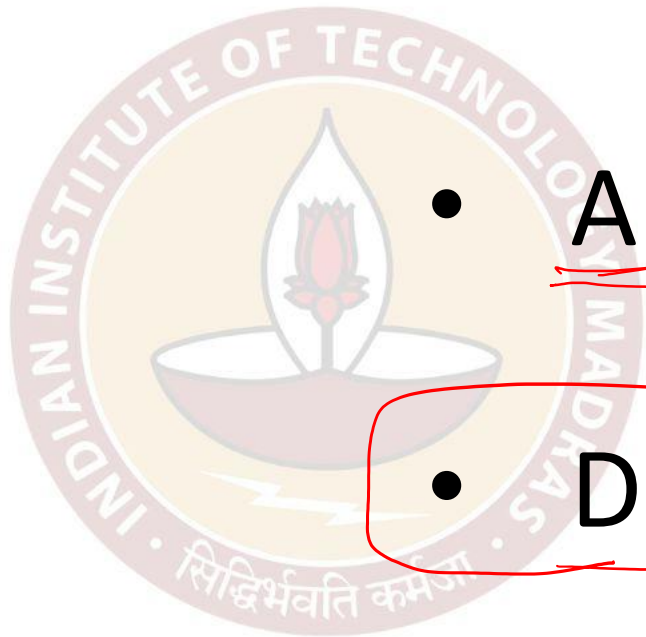
Outline:

- Professional presentation - setting ←
- **Constraints** ←
- Connecting with your audience
- Illustrations
- Summary



IIT Madras
B.S. Degree

Constraints

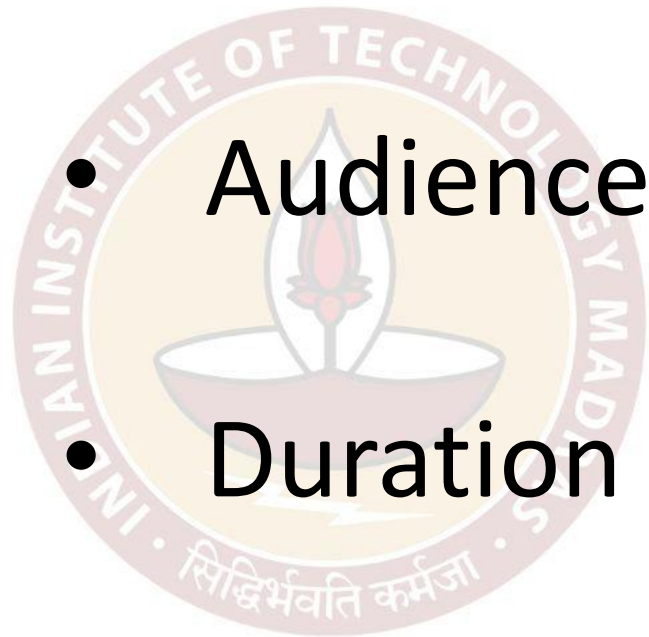


- Audience

- Duration

IIIT Madras
B.S. Degree

Constraints



- Audience: Seniority and purpose
- Duration

IIT Madras
B.S. Degree

Constraints

- Audience: Seniority and purpose
- Duration: Very important in a professional setting

Outline:

- Professional presentation - setting ✓
- Constraints ✓
- **Connecting with your audience** ←
- Illustrations
- Summary

Connecting with your audience

- Repeat a summary slide from
your previous presentation }
- Separate facts from opinions ←
- Pause to enable decisions ←

Connecting with your audience

- Use familiar formats ←
- Explain what is being shown and how to understand it }
- Special effects can be distracting

Reading from a slide

Fuel Cell Stack Development:



Short term goals:

- It is important to demonstrate scale up of single cells on test facility. Smaller 50 cm² cells often perform significantly different relative to larger cells such as 400 cm² cells. Systematic testing at a single cell level is necessary to identify the sources of such differences and to compensate for the same.
- Scale up of stacks is also a challenge. Single cells differ from a collection of cells since there is often a cell to cell variation in reactant distribution and hence current distribution. The poorest performing cell often limits the entire stack. This limitation of the stack can be effectively studied only at the stack level.
- Demonstration of stack as a stand alone unit is an important end goal. If the stack as well as its auxiliaries consume more power than the stack produces then it does not meet its objectives. Therefore the stack, its sizing and the choice of the auxiliaries should be made judiciously to accomplish the goal of net power from the stack in an efficient manner

Long term goals:

- Identification of scope for improvements in components has to be carried out since each component affects the overall efficiency and life of the stack. The catalysts, gas diffusion layers, membranes, and bipolar plates, each need to be individually improved. Standalone tests should help in the identification of improvements to the individual components.
- Design of stack to ensure ease of maintenance. During the use of the fuel cell system, with time several components are likely to degrade and fade in performance. It is necessary to develop tests to identify the state of performance of each of the individual components. Importantly, it is necessary to design the stack such that a poorly performing component can be easily removed and replaced from the stack.
- Accomplishment of load following capabilities of the stack to suit user requirement is necessary to enable the use of the stack in a real life setting. The load cycle encountered in a house is different from that encountered in an automobile. In addition load cycle in winter may be different from that in summer in a home. The Fuel cell stack and its operating scheme should be able to meet the load cycle it has to support.

Fuel Cell Stack Development:

Short term goals:

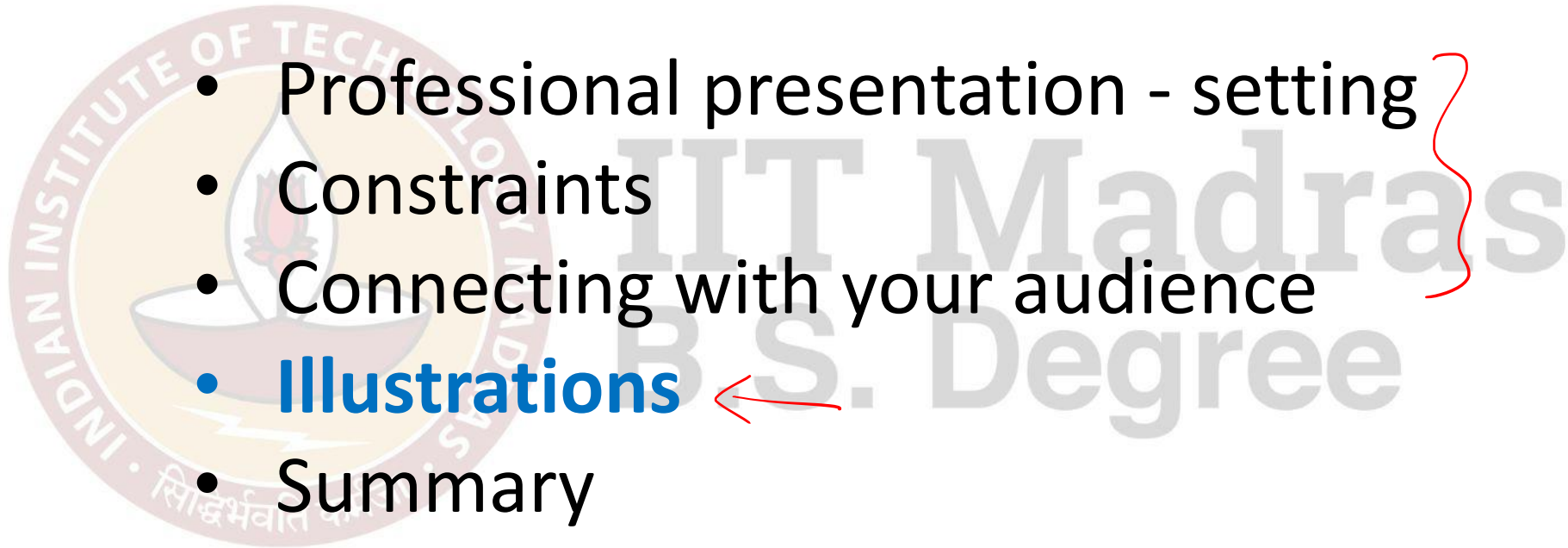
- Scale up of single cells
- Scale up of stack
- Demonstration of stack as a stand alone unit.

Long term goals:

- Scope for improvements in components
- Design for ease of maintenance.
- Load following capabilities of the stack

Outline:

- Professional presentation - setting
- Constraints
- Connecting with your audience
- **Illustrations**
- Summary



Illustration

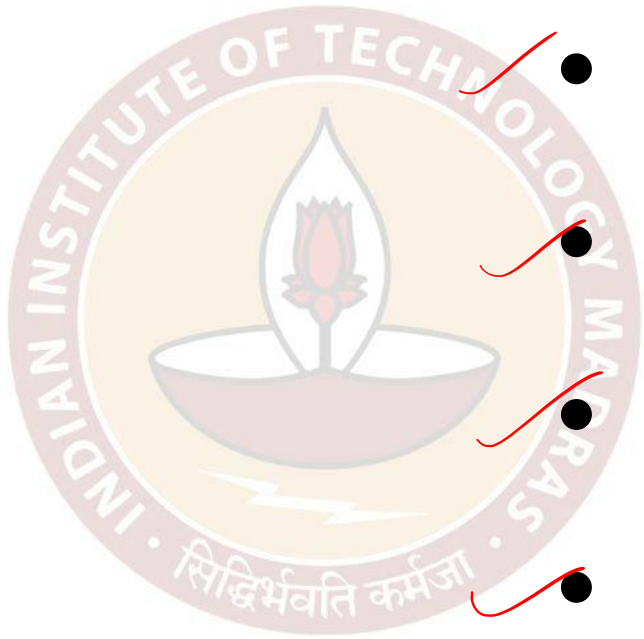


- Choosing the right type of illustration ←

- Paying attention to details ←

IIT Madras
B.S. Degree

Types of illustration



- Tables
- Graphs
- Photographs
- Drawings
- Schematics

IIT Madras
B.Sc. Degree

Table



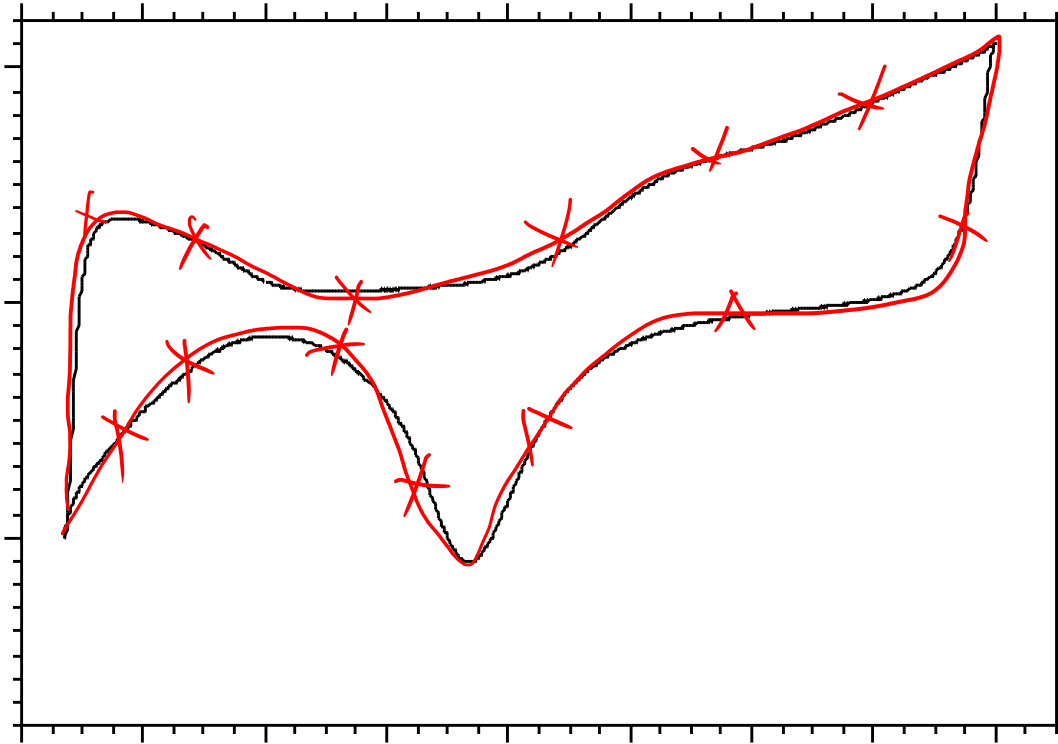
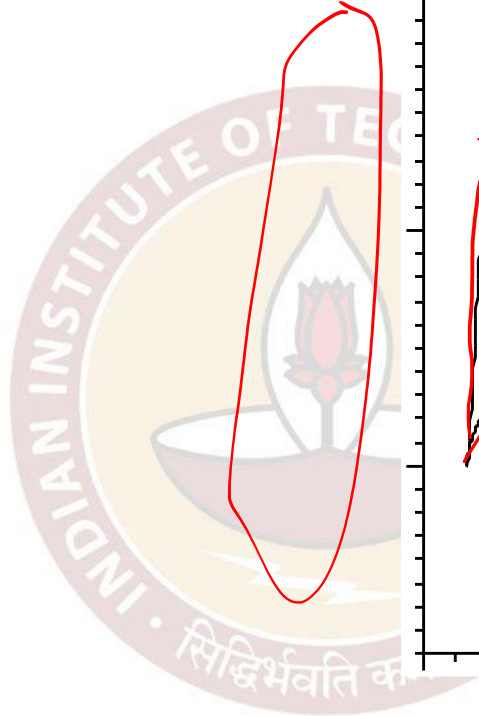
Time (Min)	First Peak position (2θ)	First Peak width	Second Peak position (2θ)	Second Peak width
Fresh	19.967134	3.203901	<u>40.93</u>	1.921001
5	20.044281	3.299972	41.039333	2.151008
10	20.015936	3.612154	41.006262	2.272935
15	20.652753	<u>4.837113</u>	<u>41.2</u>	3.345802

Table



Time (Min)	First Peak position (2θ degrees)	First Peak Width (2θ degrees)	Second Peak position (2θ degrees)	Second Peak width (2θ degrees)
Fresh	19.9	3.2	40.9	1.9
5	20.0	3.2	41.0	2.1
10	20.0	3.6	41.0	2.2
15	20.6	4.8	41.2	3.3

Graph

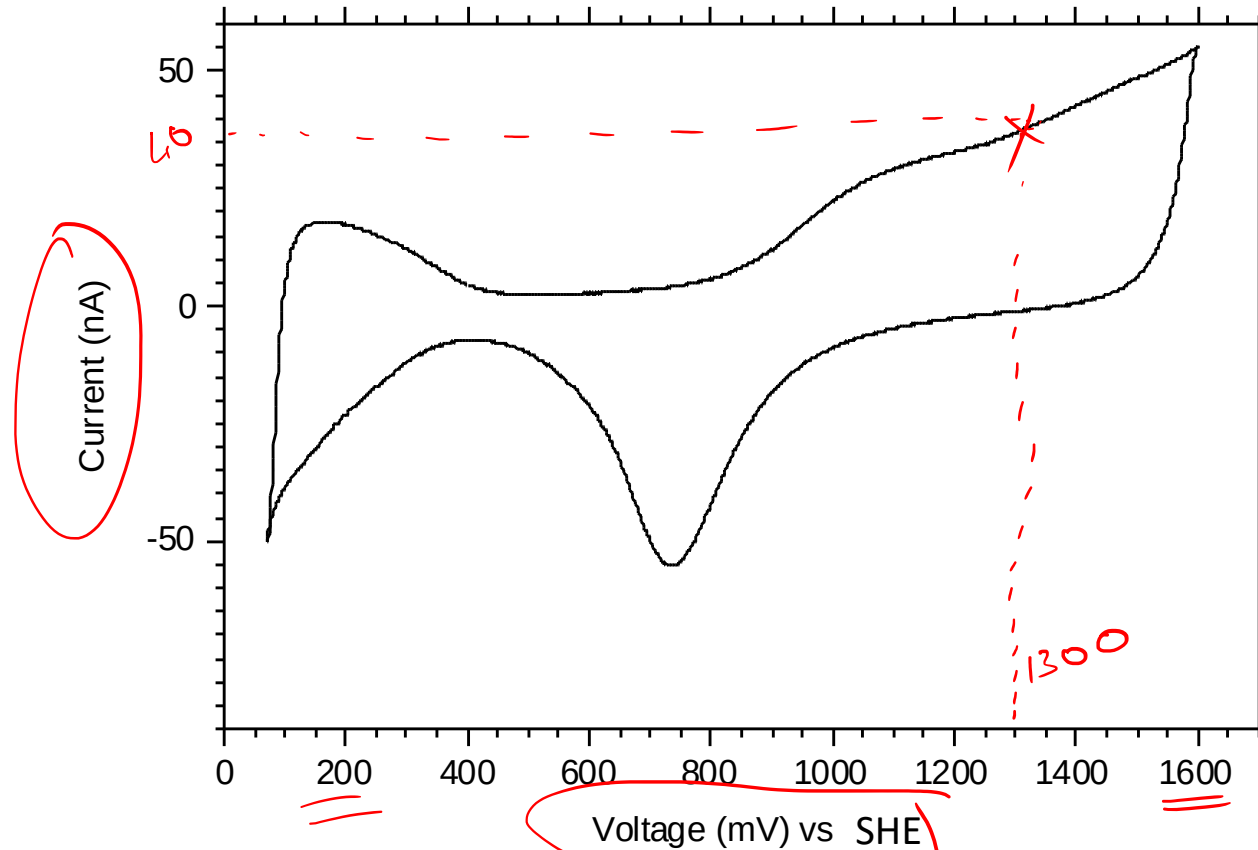


Cyclic voltammogram

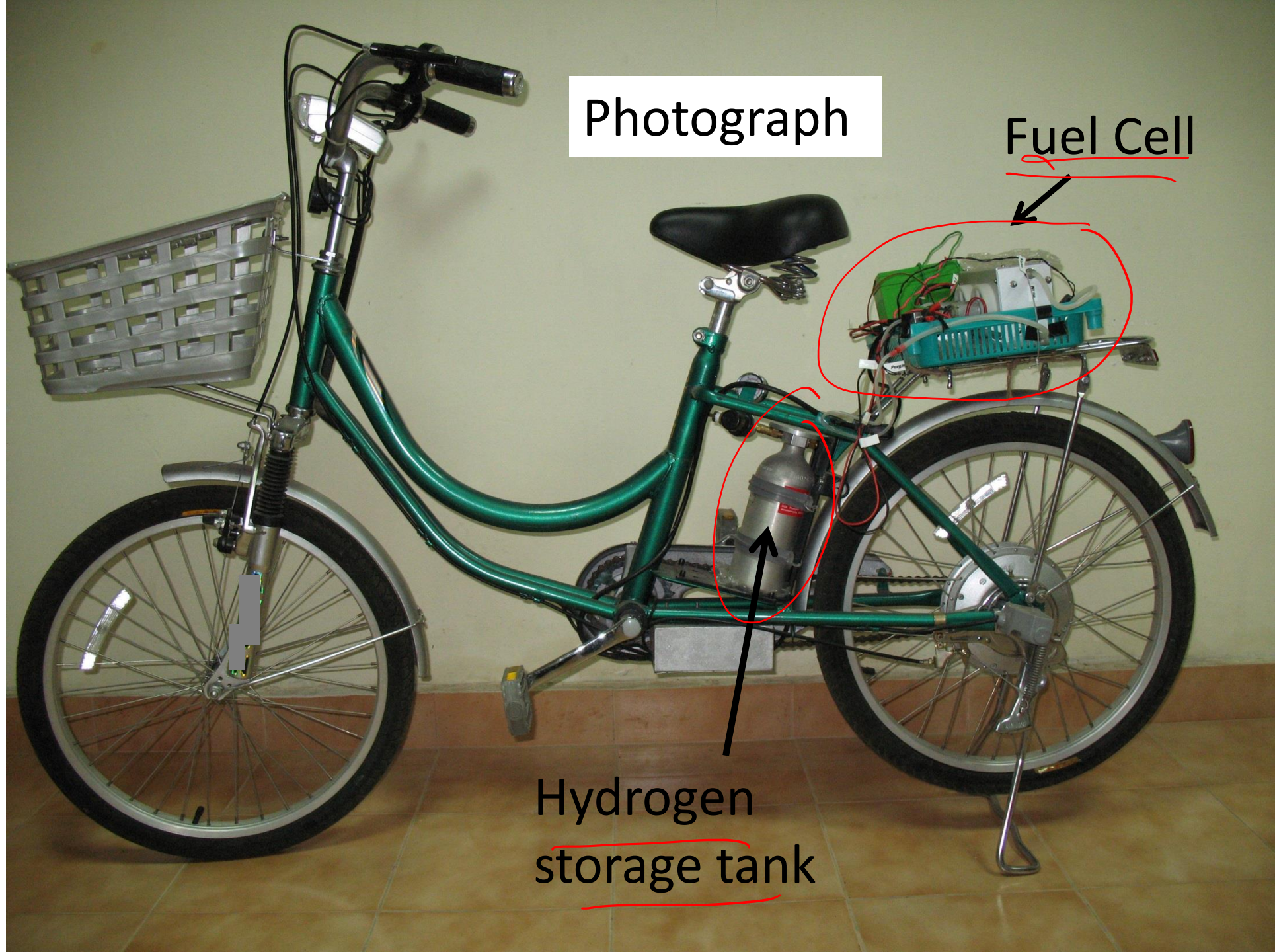


↓	
x	y
1	2
3	4
5	6
7	8
9	10
11	12
13	14
15	16

Graph



**Cyclic voltammogram from a microelectrode with a humidified recast polymer film as the electrolyte in the system and with humidified Ar flowing through the bottle.
Scan rate 200 mV/s**



Photograph

Fuel Cell

Hydrogen
storage tank



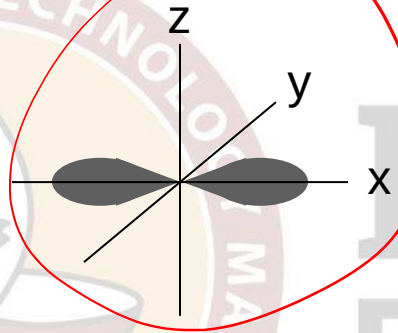


“Drawings”

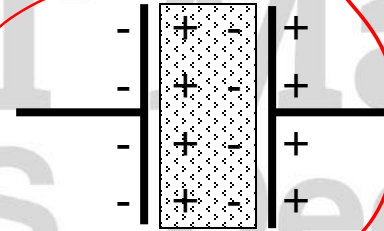
Mechanical



Chemical



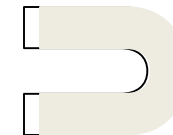
Electrical



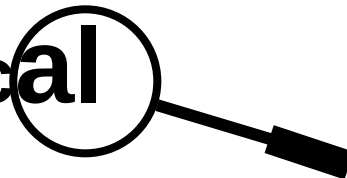
Thermal



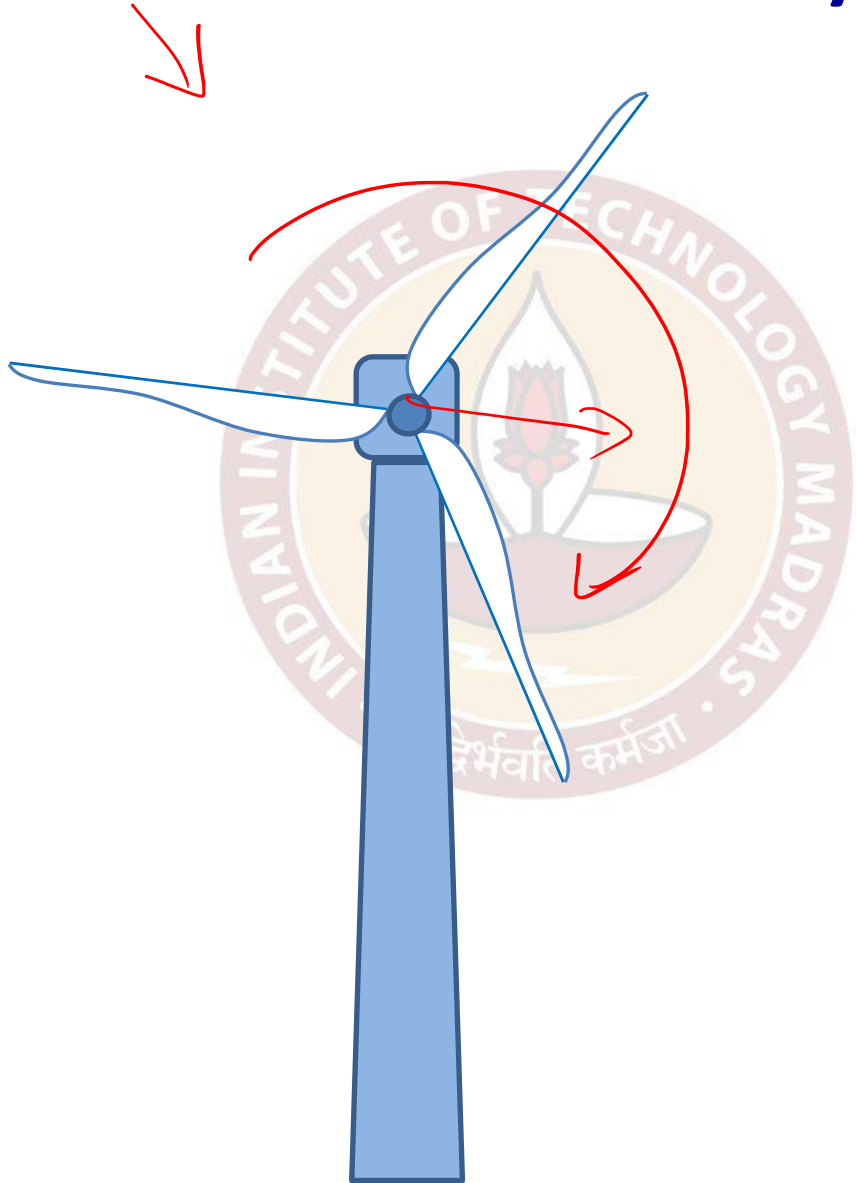
Magnetic



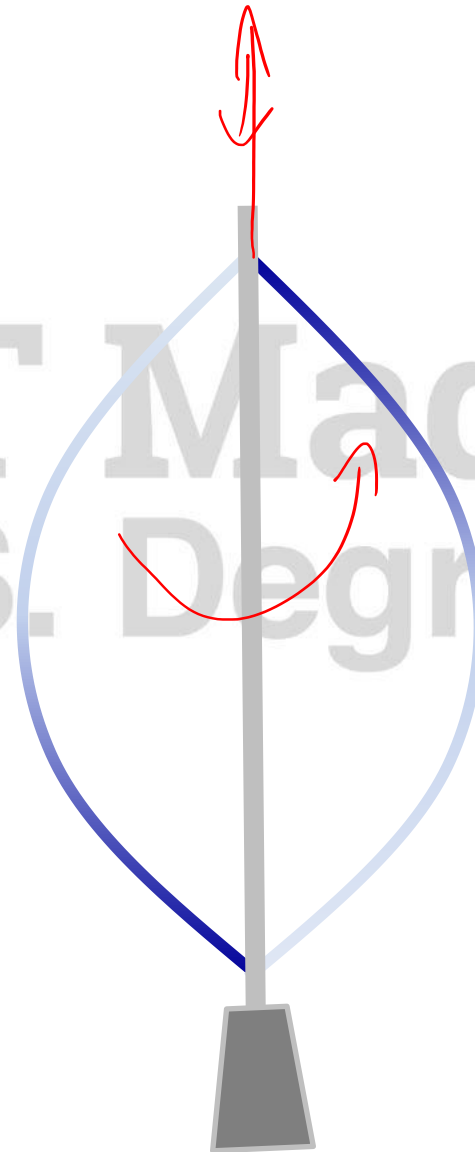
Optical



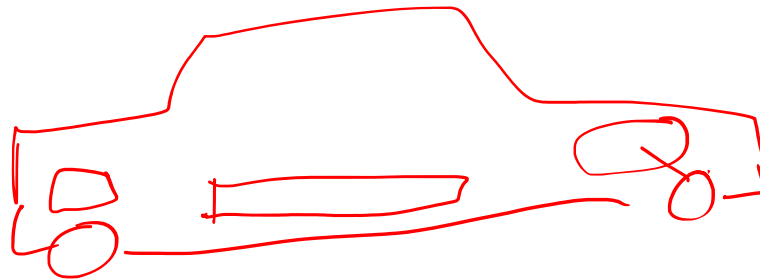
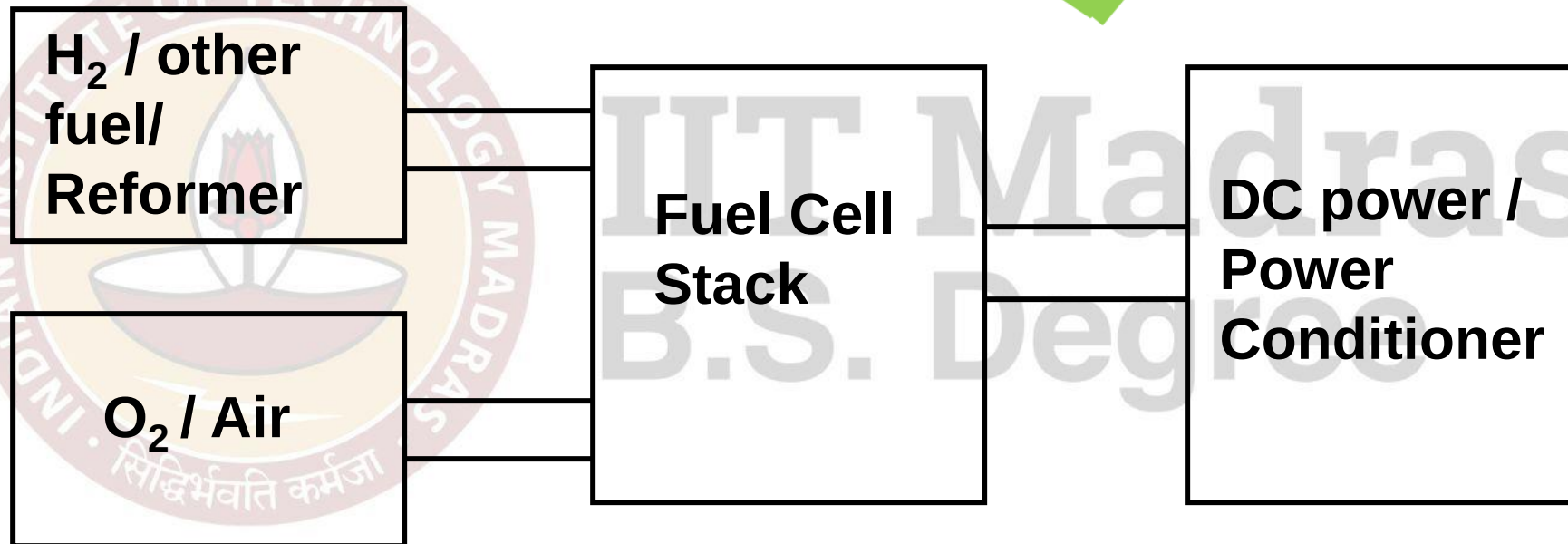
Different types of wind turbines:



IIT Madras
B.S. Degree



Schematic of a Fuel Cell system



Modulus of elasticity
Yield strength
Tensile strength
Ductility
Resilience
Toughness
Hardness

Stress

Strain

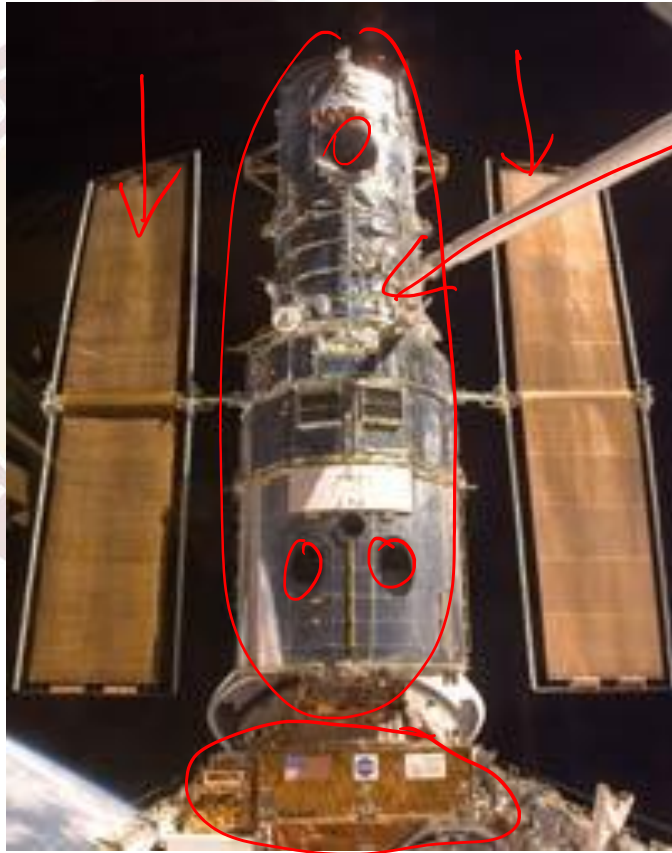


Mechanical

Properties



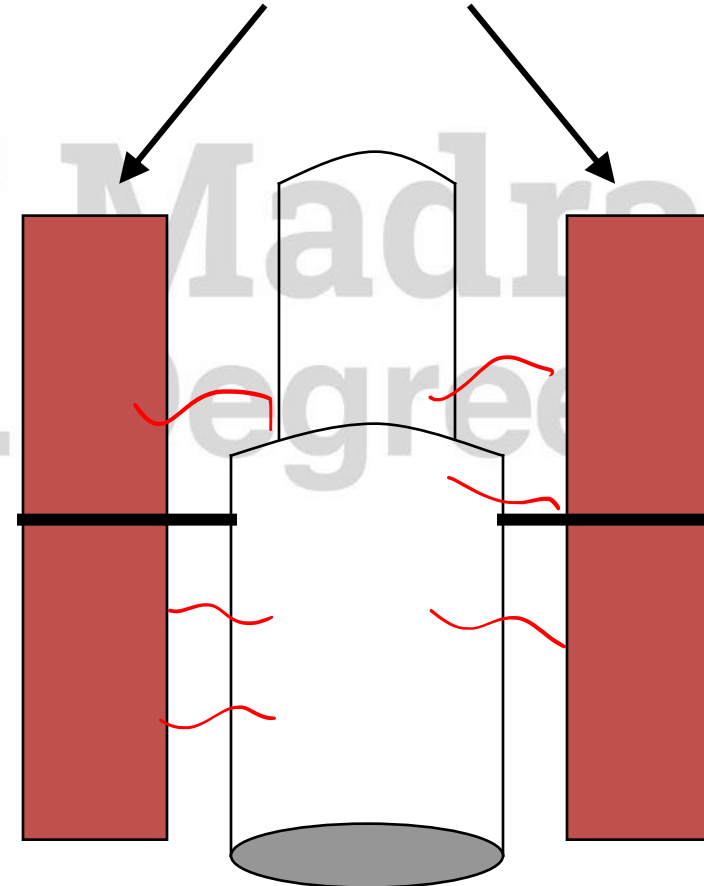
Level of complexity



Photograph

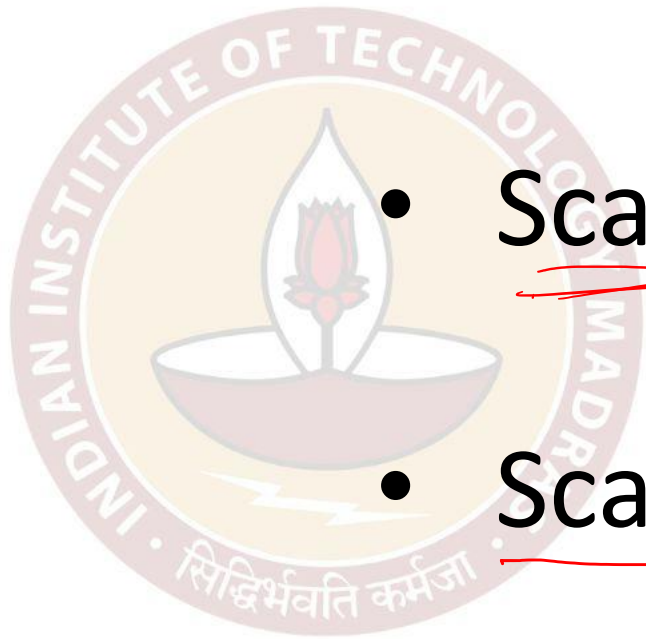


Solar panels



Drawing

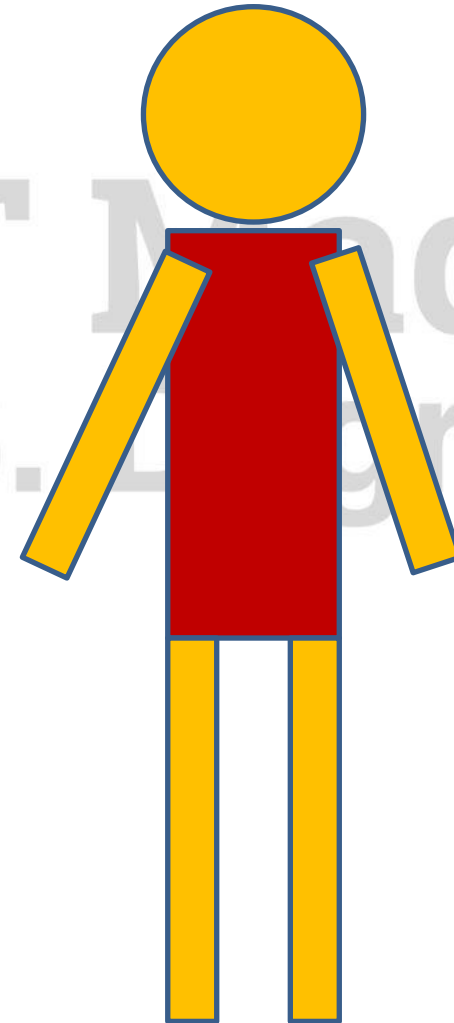
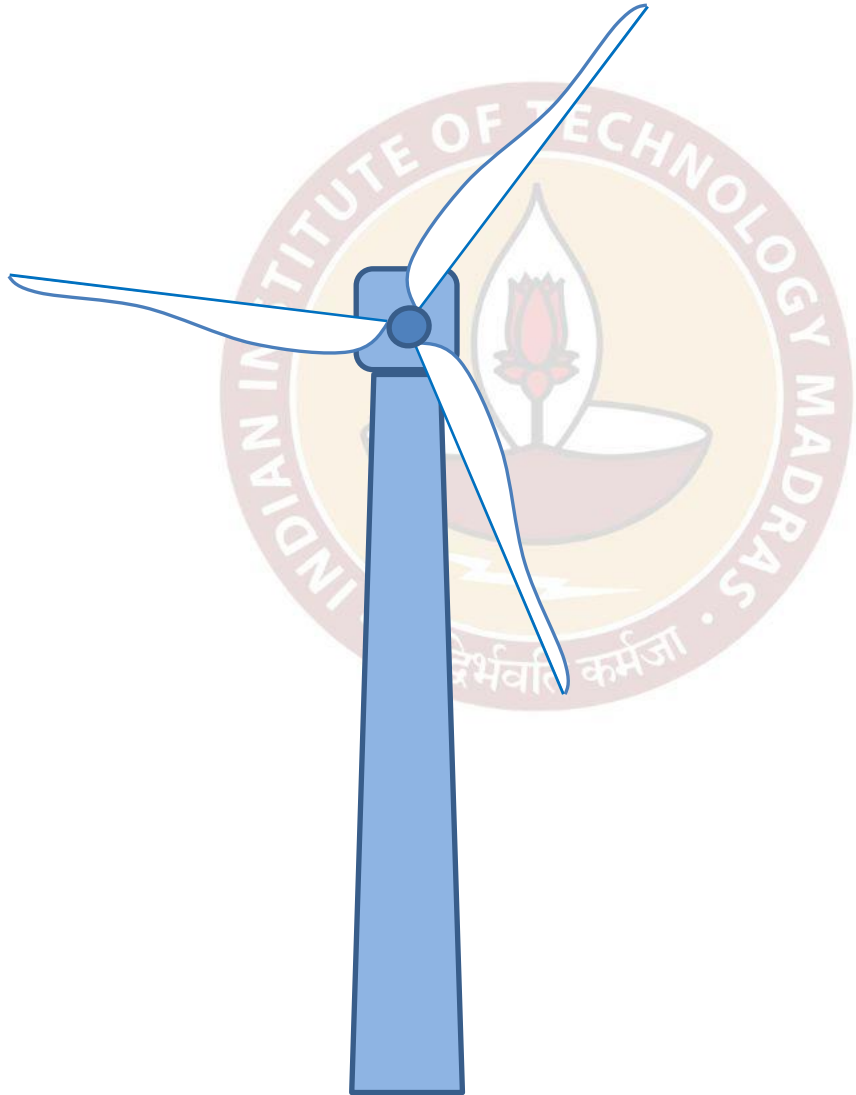
Importance of scale



- Scale of impact
- Scale of object

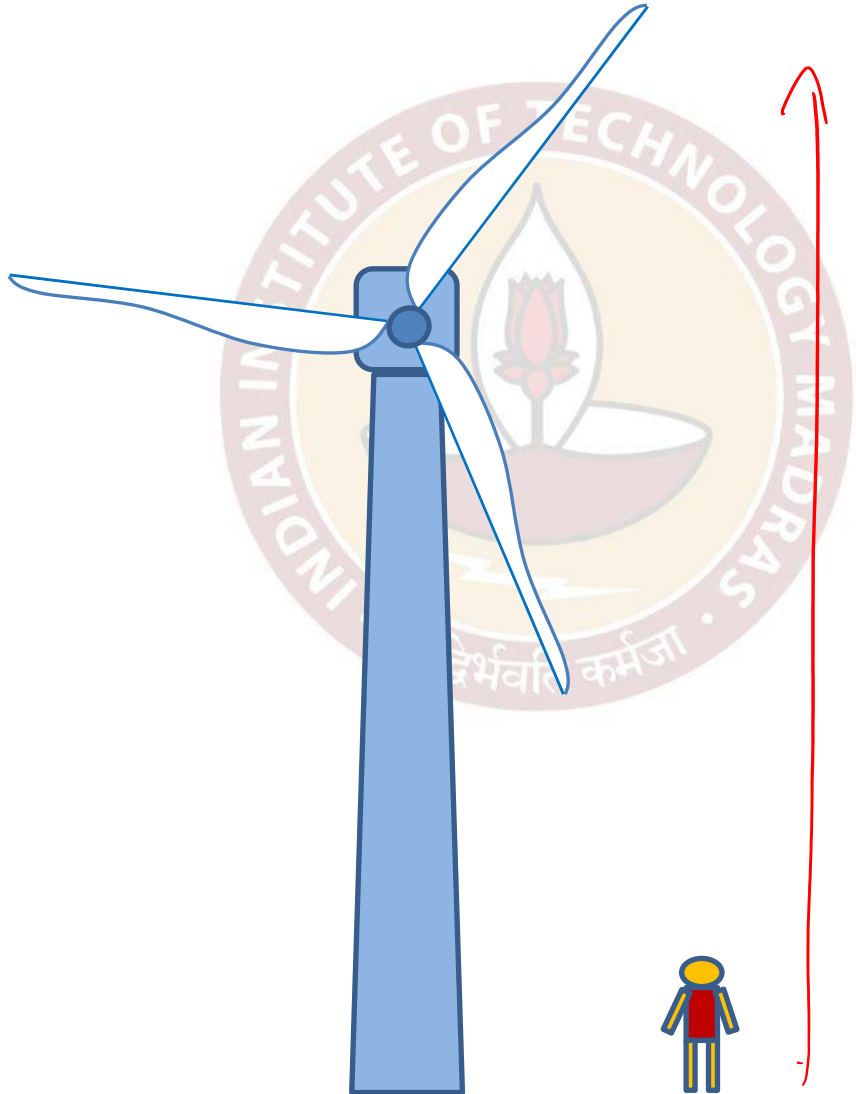
IIT Madras
B.S. Degree

Conveying a sense of scale



IIT Madras
B.S. Degree

Conveying a sense of scale



IIT Madras
B.S. Degree



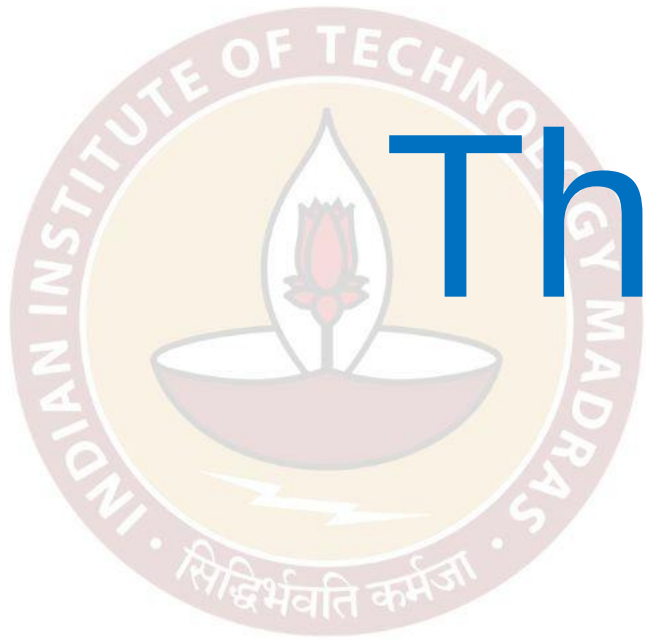
Outline:

- Professional presentation - setting
- Constraints
- Connecting with your audience
- Illustrations

• **Summary**

Summary:

- Provide an Outline ←
- Audience and duration are important constraints ←
- Connect with your audience ←
 - Tailor to audience ←
 - Orient them with previous work ←
 - Enable decisions ←
- Illustration
 - Choice of illustration
 - Pay attention to details
 - Level of complexity ←
 - Importance of scale ←
- Make a Summary



Thank you!

IIT Madras
B.S. Degree